MAKERERE UNIVERSITY

We Build For The Future

Week 3

BSE3214 – Cloud Computing and Big Data

Economics of Cloud Computing

Lecture Material

Part 3 – 15 Hours

Instructor

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Week 3

Learning Objectives

By the end of this lecture, students will be able to:

- Review cloud service models with specific focus on economic implications
- Conduct SWOT analysis and articulate clear value propositions
- Identify general risks affecting cloud economics (financial, security, operational)
- Perform cloud computing cost analysis shifting from CapEx to OpEx
- Evaluate key criteria for selecting appropriate IaaS providers
- Understand cloud standards and interoperability challenges
- Learn best practices and recommendations for successful cloud migration

Recap

Cloud Service Models

Economic Focus & Responsibility Shift

SaaS, PaaS, IaaS Overview

Brief recap of the three primary service models: Software, Platform, and Infrastructure as a Service.

Economic Differences

Variations in cost structure (subscription vs. usage) and billing granularity across models.

Management Responsibility

Shift of operational burden from user to provider, reducing internal maintenance costs.

CapEx vs. OpEx

Transition from Capital Expenditure (upfront hardware) to Operational Expenditure (pay-as-you-go).

SaaS Economics

Economic Impact of SaaS

Software as a Service Cost Benefits & Considerations

Minimal Upfront Investment

Eliminates heavy capital expenditure on software licenses and supporting hardware.

Subscription Pricing Models

Predictable monthly or annual operational expenses (OpEx) based on usage or seats.

Reduced IT Staff Overhead

Vendor manages maintenance, updates, and patches, freeing internal IT resources.

Rapid Return on Investment (ROI)

Immediate access to standard business applications accelerates time-to-value.

Hidden Costs

Potential expenses for customization, complex integrations, and data egress fees.

Economics

Economic Impact of PaaS

Development Efficiency & Cost Structures

Subscription & Usage Model

Platform subscription fees combined with pay-for-usage charges based on consumption.

Time-to-Market Reduction

Faster development cycles significantly reduce overall project costs and accelerate revenue generation.

Platform Lock-in Risks

Potential long-term costs associated with proprietary technologies affecting future switching capabilities.

Managed Middleware Savings

Cost efficiencies gained from provider-managed middleware, runtime environments, and OS maintenance.

Economics

Economic Impact of IaaS

Infrastructure as a Service Cost Dynamics

Pay-as-you-go Model

Cost efficiency through pay-as-you-go pricing for virtualized compute, storage, and network resources.

CapEx to OpEx Shift

Enables significant financial shift from capital expenditure (hardware) to operational expenditure (services).

Management Costs

Requires ongoing internal management and operational costs for OS patching, security, and configuration.

Cost Overrun Risks

Significant potential for cost overruns without proper monitoring, governance, and resource management.

SWOT Analysis

Cloud Strengths

Strengths

Internal Positive Factors

- **Scalability** reduces over-provisioning costs (pay for what you use)
- **Reduced infrastructure management** frees up IT resources
- **Global reach & elasticity** significantly increase business agility
- **Innovation enablement** through rapid service provisioning

SWOT Analysis

Cloud Weaknesses

Weaknesses

Internal Negative Factors

- **Vendor lock-in risks** due to proprietary platforms and APIs
- **Possible hidden or variable costs** (egress, API calls, storage tiers)
- **Limited customization** in standard SaaS and PaaS offerings
- **Security & compliance concerns** leading to potential fines

SWOT Analysis

Cloud Opportunities

Opportunities

External Positive Factors

- **New business models** enabled by advanced cloud capabilities
- **Integration with AI, Big Data, IoT** reducing operational costs
- **Hybrid and multi-cloud strategies** optimize costs and control
- **Outsourcing non-core IT** reduces focus burden on internal teams

SWOT Analysis

Cloud Threats

Threats

External Negative Factors

- **Increasing competition** driving pricing pressure
- **Regulatory changes** increasing compliance costs
- **Data breaches and outages** impacting reputation and finances
- **Cloud market consolidation** may reduce choices and leverage

Summary

Value Proposition of Cloud Computing

Why organizations move to the cloud: Economic & Strategic Benefits

Cost Savings

Significant reduction in hardware, maintenance, and power expenses.

Operational Agility

Rapid flexibility to respond to market changes and business needs.

Scalability Matching Demand

Prevents overcapacity spend; pay only for what you actually use.

Innovation Enabler

Rapid provisioning allows faster experimentation and time-to-market.

*"Cloud computing shifts IT from a **cost center** to a **strategic enabler**."*

Overview

Risk Categories

Major areas of concern when adopting cloud services

Category 1

Financial Risks

Challenges related to cost management, unpredictable billing, over-provisioning, and hidden expenses associated with cloud resource consumption.

Category 2

Security & Compliance

Concerns regarding data breaches, identity theft, regulatory compliance failures (GDPR, HIPAA), and data sovereignty issues.

Category 3

Vendor & Technology

Risks of vendor lock-in, proprietary technology dependencies, service discontinuation, and lack of interoperability between providers.

Category 4

Operational Risks

Issues affecting daily operations including service outages, performance degradation, skill gaps, and complexity in multi-cloud management.

Risk Category

Financial Risks

Cost Management Challenges in Cloud Environments

Unpredictable Usage & Billing

Dynamic resource consumption can lead to unexpected bill spikes ("Bill Shock"), especially with auto-scaling services.

Provisioning Balance

Over-provisioning wastes budget on idle resources, while under-provisioning risks performance degradation and lost revenue.

Hidden Costs

Accumulation of often-overlooked charges including data transfer (egress) fees, backup storage, premium support, and advanced monitoring.

Premium SLAs

Higher availability guarantees often require expensive multi-region architectures or enterprise-tier support plans.

Risks

Security and Compliance Risks

Critical Factors Affecting Cloud Economics & Governance

Breaches & Financial Loss

Data breaches can lead to significant financial loss, heavy regulatory fines, and long-term reputational damage.

Data Sovereignty & Complexity

Regional data residency laws (like GDPR) increase architectural complexity and cost for multi-region deployments.

Auditing & Reporting Overhead

Meeting stringent compliance standards requires continuous monitoring, reporting tools, and dedicated staff resources.

Risk Analysis

Vendor and Technology Risks

Strategic Dependencies and Long-term Challenges

Proprietary Lock-in

Dependency on vendor-specific APIs, data formats, and unique platform features creates high switching costs, making it difficult to migrate to other providers.

Product Discontinuation

Risk of providers retiring specific services, versions, or products (deprecation), forcing customers to undertake unplanned re-architecture or migration efforts.

Integration Complexity

Significant challenges in bridging modern cloud-native services with existing on-premises legacy systems, increasing technical debt and future migration effort.

Risks

Operational Risks

Disruptions, Skill Gaps, and Management Challenges

Service Outages & Downtime

Outages resulting in business disruption and financial loss. Even with high availability SLAs, cloud provider failures can halt critical operations.

Staff Skill Gaps

Lack of specialized cloud expertise increasing operational cost. Organizations may face higher labor expenses to hire or train qualified personnel.

Multi-Cloud Complexity

Complex multi-cloud management overhead. Balancing workloads across different providers adds layers of operational difficulty and requires advanced tooling.

Analysis

Cost Components in Cloud

Breaking Down Cloud Economics & Expenditure

Compute, Storage & Network

Core infrastructure costs based on instance types, storage tiers, and data transfer (egress/ingress).

Licensing & Software Fees

Additional costs for OS licenses (Windows/RHEL), database engines, and marketplace software subscriptions.

Management & Support

Expenses for technical support plans (Business/Enterprise), training staff, and managed services fees.

Security & Compliance Overhead

Costs for dedicated security tools, auditing, encryption services, and compliance reporting mechanisms.

Analysis

Cost Comparison: Cloud vs On-Premises

Comparing financial models and operational impacts

Cost Type	On-Premises IT	Cloud Computing
Capital Expense (CapEx)	**↑ High Upfront Investment** Servers, storage hardware, networking gear, software licenses, cooling systems bought in advance.	**↓ Minimal / Zero Upfront** No hardware purchase required. Infrastructure is accessed as a service.
Operational Expense (OpEx)	**Ongoing Fixed Costs** Electricity, cooling, physical security, IT staff salaries, maintenance contracts.	**Variable Subscription** Pay-as-you-go model based on actual consumption. Predictable monthly billing.
Maintenance	**High Burden** Manual hardware refresh cycles (3-5 years), OS patching, upgrades managed internally.	**Provider Managed** Hardware maintenance, virtualization layer, and physical security handled by provider.
Scalability Costs	**Expensive & Slow** Must buy for peak capacity. Over-provisioning leads to wasted resources.	**Efficient & Elastic** Scale up/down instantly. Pay only for what you use, eliminating idle capacity cost.

Strategies

Cost Optimization Strategies

Proactive approaches to manage and reduce cloud spending

Right-Sizing & Right-Scaling

Match resource types/sizes to workload needs and implement auto-scaling policies.

Reserved Instances & Savings Plans

Commit to usage for 1-3 years to secure significant discounts vs. on-demand rates.

Spot / Preemptible Instances

Utilize spare capacity at up to 90% discount for fault-tolerant, interruptible workloads.

Monitoring & Alerting

Set budget alerts and detect anomalies early to prevent unexpected bill surprises.

Storage Tiering

Move infrequently accessed data to cheaper cold storage classes automatically.

Cleanup Unused Resources

Identify and remove orphaned volumes, unattached IPs, and idle load balancers.

*"Effective **FinOps** requires continuous monitoring, analysis, and adjustment."*

Tools

Tools for Cost Analysis

Software solutions for monitoring, analyzing, and optimizing cloud spend

Native Provider Tools

Cloud Provider Calculators

Official tools like **AWS Cost Explorer**, **Azure Pricing Calculator**, and **Google Cloud Billing**. Best for estimating costs before deployment and analyzing historical spend data directly within the console.

Third-Party SaaS

Enterprise Cost Management

Specialized platforms like **Cloudability (Apptio)** and **CloudHealth (VMware)**. These offer multi-cloud visibility, advanced budgeting, anomaly detection, and automated optimization recommendations for large organizations.

Open Source

Kubernetes Cost Management

Tools like **Kubecost** designed specifically for containerized environments. Provides granular cost visibility by namespace, deployment, or service, helping allocate shared cluster costs accurately.

Selection Criteria

Pricing & Billing Models

Evaluating IaaS providers based on cost structure and transparency

Core Pricing Models

Pay-As-You-Go (On-Demand)

Standard model with no upfront commitment. Ideal for spiky workloads, short-term experiments, or unpredictable traffic patterns. Highest per-hour cost but maximum flexibility.

Reserved Instances / Savings Plans

Commitment to use specific resources for 1-3 years in exchange for significant discounts (up to 72%). Best for steady-state, predictable production workloads.

Spot Instances

Bid on unused capacity for massive discounts (up to 90%). Instances can be reclaimed with short notice. Suitable for fault-tolerant, interruptible batch jobs.

Billing Considerations

Billing Granularity

Does the provider bill per second, per minute, or per hour? Per-second billing saves money for short-lived tasks like serverless functions or auto-scaling groups.

Hidden Fees & Costs

Look beyond compute costs. Check pricing for data egress (transfer out), API request volume, storage operations (I/O), and snapshot retention.

Enterprise Billing Features

Availability of consolidated billing for multiple accounts, volume discounts, cost allocation tags, and budget alert capabilities for governance.

Provider Evaluation

Performance & SLAs

Key technical metrics and service commitments for reliable operations

Performance Metrics

Compute Resources

Evaluate vCPU/RAM ratios, processor generations, and availability of specialized hardware (GPUs/TPUs) for high-performance workloads.

Storage I/O (IOPS)

Assess disk throughput and IOPS guarantees. Understand differences between HDD, standard SSD, and provisioned IOPS SSD performance tiers.

Network Throughput

Check bandwidth limits per instance type and latency between availability zones. Ensure sufficient capacity for data-intensive applications.

Service Level Agreements

Uptime Guarantees

Review availability commitments (e.g., 99.9% vs 99.99%). Understand definition of "uptime" and maintenance windows exclusions.

Penalties & Credits

Examine compensation mechanisms for SLA breaches. Typically provided as service credits rather than cash refunds, tiered by outage severity.

Support Quality

Evaluate support tier responsiveness, channel availability (phone/chat/ticket), and technical expertise levels included in base vs. premium plans.

Selection Criteria

Security & Compliance

Evaluating provider capabilities in data protection and regulatory adherence

Security Standards

ISO 27001 Certification

International standard for Information Security Management Systems (ISMS). Ensures the provider follows rigorous best practices for securing infrastructure and data.

SOC 2 Type II

Audited report focusing on non-financial reporting controls relevant to security, availability, processing integrity, confidentiality, and privacy.

HIPAA Compliance

Essential for healthcare-related workloads. Ensures the provider can sign Business Associate Agreements (BAAs) to safeguard Protected Health Information (PHI).

Compliance Features

Data Encryption & Isolation

Look for strong encryption options for data at rest and in transit. Verify Virtual Private Cloud (VPC) capabilities for robust network isolation and segmentation.

Regional Regulation (GDPR)

Crucial for handling EU citizen data. Ensure the provider offers data residency options and tools to manage data sovereignty requirements compliant with GDPR.

Consumer Privacy (CCPA)

Compliance with the California Consumer Privacy Act. Provider tools should assist in data discovery and subject rights request fulfillment.

Selection Criteria

Ecosystem & Tools

Assessing platform maturity, integrations, and operational tooling

Platform Ecosystem

Integrations & Connectors

Availability of pre-built integrations with essential enterprise systems (ERP, CRM) and seamless connectivity with on-premises infrastructure for hybrid deployments.

Partner Network

Strength of the provider's consulting (MSP) and technology partner network. Access to certified experts for migration, optimization, and managed services.

Marketplace Solutions

Richness of the cloud marketplace offering third-party software (firewalls, databases, analytics tools) that can be deployed with one-click and billed through the cloud invoice.

Operational Tooling

Automation & DevOps

Support for Infrastructure as Code (Terraform, native templates), robust SDKs/CLIs, and integration with CI/CD pipelines for automated deployment and management.

Monitoring & Observability

Native tools for detailed performance monitoring, log aggregation, and tracing. Capabilities for setting alarms and creating custom dashboards for visibility.

Management & Governance

Tools for centralized management of resources, policy enforcement, tagging strategies, and identity management integration to maintain control at scale.

Standards

Importance of Cloud Standards

Why standardization is crucial for long-term cloud success and flexibility

Avoid Vendor Lock-In

Prevent dependency on proprietary APIs and data formats for future flexibility.

Enable Portability

Ensure seamless movement of workloads and data across different cloud environments.

Facilitate Hybrid / Multi-Cloud

Streamline integration between on-premises and multiple public cloud providers.

Promote Consistency

Unified approach to security, compliance, and governance across all platforms.

*"Standards are the **common language** that allows diverse cloud systems to communicate and cooperate."*

Standards

Key Standards and Frameworks

Essential protocols and guidelines ensuring interoperability and security

Interface Standard

OCCI (Open Cloud Computing Interface)

A set of open specifications for cloud management interfaces. It provides a RESTful protocol and API for all kinds of management tasks, enabling interoperability across different IaaS providers.

Infrastructure Mgmt

CIMI (Cloud Infrastructure Management Interface)

Developed by DMTF, this standard simplifies management of cloud infrastructure. It standardizes interactions between cloud consumers and providers for managing resources like machines, volumes, and networks.

Orchestration

TOSCA (Topology & Orchestration)

An OASIS standard language to describe cloud application topology and orchestration. It enables portable automated deployment and management of applications across different cloud platforms.

Security Best Practices

Cloud Security Alliance (CSA)

Not a protocol standard but a crucial framework of best practices. The CSA Cloud Controls Matrix (CCM) provides a fundamental security baseline for cloud computing assurance and compliance.

Challenges

Interoperability Challenges

Barriers to Seamless Cloud Integration

Proprietary APIs & Features

Vendor-specific implementations create lock-in and complicate cross-cloud development.

Data & Protocol Incompatibility

Differences in data formats and communication protocols hinder seamless data exchange.

Identity & Access Management

Inconsistent IAM models (RBAC vs. ABAC) make unified security administration difficult.

Distributed Governance

Complexity in maintaining consistent compliance, logging, and policies across multiple providers.

Migration

Pre-Migration Planning

Critical steps before moving workloads to the cloud

1

Inventory & Audit

Map all workloads, dependencies, and assess "data gravity" to understand migration complexity.

2

Define Objectives

Clarify goals: Cost reduction, scalability, resilience, or security enhancement.

3

Assess Compliance

Review regulatory requirements and data sovereignty laws (GDPR, local acts) early.

4

Select Models

Choose the right service (IaaS/PaaS/SaaS) and deployment model (Public/Private/Hybrid).

5

Business Case

Build ROI projections and define clear success metrics (KPIs) for the migration.

Recommendations

Migration Best Practices

Key strategies for successful and efficient cloud migration

Start Small

Begin with pilot projects or non-critical workloads to validate the approach.

Leverage Cloud-Native Tech

Utilize containers, microservices, and serverless for greater efficiency.

Automate Everything

Use Infrastructure as Code (IaC) tools for consistent, repeatable deployment.

Continuous Optimization

Monitor performance and costs post-migration to refine and improve.

*"Successful migration is not just lifting and shifting, but **transforming** for the cloud."*

Governance

Post-Migration Optimization & Governance

Sustaining cloud value through continuous improvement and control

Implement Comprehensive Monitoring

Track cost, security, and performance metrics continuously.

Train Staff on Best Practices

Upskill team on cloud management, FinOps, and security protocols.

Establish Cloud Governance

Create frameworks for policy enforcement, compliance, and access control.

Plan for Future Innovation

Prepare roadmap for evolving cloud capabilities and new technologies.

"Governance is not about restriction, but about enabling ***safe velocity****."*

Monitor → Optimize → **Govern**